Homework 3: Chapters 2 & 3

Mathematical Models

Author

Paul Beggs
BeggsPA@Hendrix.edu

Instructor

Dr. Christopher Camfield, Ph.D.

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Chapter 2

9. Reconsider the newspaper problem of Exercise 7, but now look at the newspaper's business expenses. The current weekly business expenses for the paper are as follows: \$80,000 for the editorial department (news, features, editorials); \$30,000 for the sales department (advertising); \$30,000 for the circulation department; and \$60,000 in fixed costs (mortgage, utilities, maintenance). The new management is considering cuts in the editorial department. It is estimated that the paper can operate with a minimum of a \$40,000/week editorial budget. Reducing the editorial budget will save money, but it will also affect the quality of the paper. Experience in other markets suggests that the paper will lose 2% of its subscribers and 1% of its advertisers for every 10% cut in the editorial budget. Management is also considering an increase in the sales budget. Recently the management of another paper in a similar market expanded its advertising sales budget by 20%. The result was a 15% increase in advertisements. The sales budget may be increased to as much as \$50,000/week, but the overall budget for business expenses will not be increased beyond the current level of \$200,000/week.

(a) Find the editorial and sales budget listings that maximize profit. Assume that the subscription price remains at \$1.50/week, and the advertising price stays at \$250/page. Use the five-step method, and model as a constrained optimization problem. Solve using the method of Lagrange multipliers. Python Code

Solution. We find that the optimal way to maximize profit is to maximize the sales budget to its limit, \$50,000, and to reduce the editorial budget to as low as possible. This conclusion comes from the profit function:

$$P(x, y) = -0.5775x + 1.45y + 10200,$$

where x and y are the editorial and sales budgets, respectively. We see that for every additional dollar spent on the editorial budget, the company actually decreases profit rather than increasing it. Thus, the new management's idea of cutting the editorial budget to \$40,000 per week is a valid plan (while also keeping the sales budget locked at \$50,000), and will net a weekly profit of \$59600.

(c) Draw a graph of the feasible region for this problem. Indicate the location of the optimal solution on this graph. Which of the constraints are binding at the optimal solution? How is this related to the shadow prices? Feasible Region

Solution. As mentioned in part (a), the sales constraint is the most binding. Regarding shadow prices, we find that by looking at lambda in our Lagrange multiplier equation, we don't even rely upon the editorial budget at all. So, by just looking at the lambda value of 1.45, we know that for every dollar spent on sales, we would increase the profit by that amount.

Chapter 3

2. Reconsider the pig problem of Example 1.1, but now suppose that the weight of the pig after t days is $w = 800/(1 + 3e^{-t/30})$ lbs.

(a) Show that the pig is gaining about 5 lbs/day at $t = 0$. What happens as t increases? Python Code

Solution. To show that the pig is gaining 5 pounds per day at $t = 0$, we find the derivative of w, $w'(t)$, and evaluate $w'(0)$. We find that $w'(0) = 5$. Using Newton's method, we find that as t increases, the growth rate of the pig slowly increases, until we hit an inflection point at $t \approx 32.96$, where the growth rate tops out at $6\frac{2}{3}$, and decreases thereafter. Then, if we assume that the pig will live forever (i.e., as $t \to \infty$), we find

$$\lim_{t\to\infty} \frac{800}{1 + 3e^{-t/30}} = \frac{800}{1 + 3(0)} = 800 \text{ lbs}$$

is the largest that the pig can get.

(b) Find the optimal time to sell the pig. Use the five-step approach, and model as a one–variable optimization problem. Python Code

Solution. Given the new function to calculate the weight of the pig with old parameters holding, the best time to sell the pig is at 12.33 days. Selling at this time yields a net profit of \$135.42.

10. Reconsider the newspaper problem of Exercise 7 in Chapter 2, but now suppose that we choose to maximize our profit margin (profit as a percentage of revenue). Assume that our business expenses remain fixed at \$200,000 per week. Python Code

(b) Let $z = f(x, y)$ denote the objective function you obtained in part (a). Use a computer algebra system to determine $F = \partial f/\partial x$ and $G = \partial f/\partial y$. Then determine $\partial F/\partial x$, $\partial F/\partial y$, $\partial G/\partial x$, and $\partial G/\partial y$.

Solution. $F = -100000x - 10y + 157500$; $G = -10x - 1y + 475$; $\partial F/\partial x = -100000$; $\partial F/\partial y = -10$; $\partial G/\partial x = -10$; and $\partial G/\partial y = -1$.

(c) Use Newton's method in two variables to obtain a precise answer to the question in part (a). Use the approximate solution from part (a) as your initial estimate. The required derivatives were calculated in part (b). Python Code

Solution. Using Newton's method with initial guesses of \$1.50 and \$350 per subscription and ad price, respectively, we find the optimal prices to be approximately \$1.53 and \$459.71, respectively.

```
# Variables
x = var('x') # editorial budget
y = var('y') # sales budget

# Parameters
x_0    = 80_000  # current editorial budget ($ / week)
y_0    = 30_000  # current sales budget ($ / week)
subs_0 = 80_000  # current subscribers (subscriber / week)
ads_0  = 350     # current ads (pages / week)

d_p_ad  = 250     # dollars per advertisement
d_p_pap = 1.50    # dollars per paper (per subscriber)

cir_budget = 30_000
fixed_cost = 60_000

x_cut_rate = .1 # the budget cut rate
y_inc_rate = .2 # sales increase rate

x_sub_loss = .02 # expected loss of subscribers (per 10% cut)
x_ads_loss = .01 # expected loss of ads (per 10% cut)
y_adv_inc  = .15 # rate increase in advertisements (per 20% budget increase)

# subs change per ad page change
slope_ads_subs = 1000 / 50

# pages per dollar spent on editorial
slope_x_ads  = (ads_0 * x_ads_loss) / (x_0 * x_cut_rate)

# pages per dollar spent on sales
slope_y_ads  = (ads_0 * y_adv_inc) / (y_0 * y_inc_rate)

# sub loss per editorial $ loss
slope_x_subs = (subs_0 * x_sub_loss) / (x_0 * x_cut_rate)

# Equations
A(x,y) = ads_0 + slope_x_ads*(x - x_0) + slope_y_ads*(y - y_0)
S(x,y) = subs_0 + slope_x_subs*(x - x_0) + slope_ads_subs*(A(x,y) - ads_0)
R(x,y) = d_p_pap * S(x,y) + d_p_ad * A(x,y)
C(x,y) = x + y + cir_budget + fixed_cost
P(x,y) = R(x,y) - C(x,y)

# Lagrange multipliers
lam = var('lam')
```

```python

g(x,y) = y
k = 50000

sales_const = solve([
    P.derivative(y) == lam * g.derivative(y),
    g == k
], x, y, lam)

print(sales_const)
print(P(40000, 50000))
```

Listing 1: Exercise 1 Code

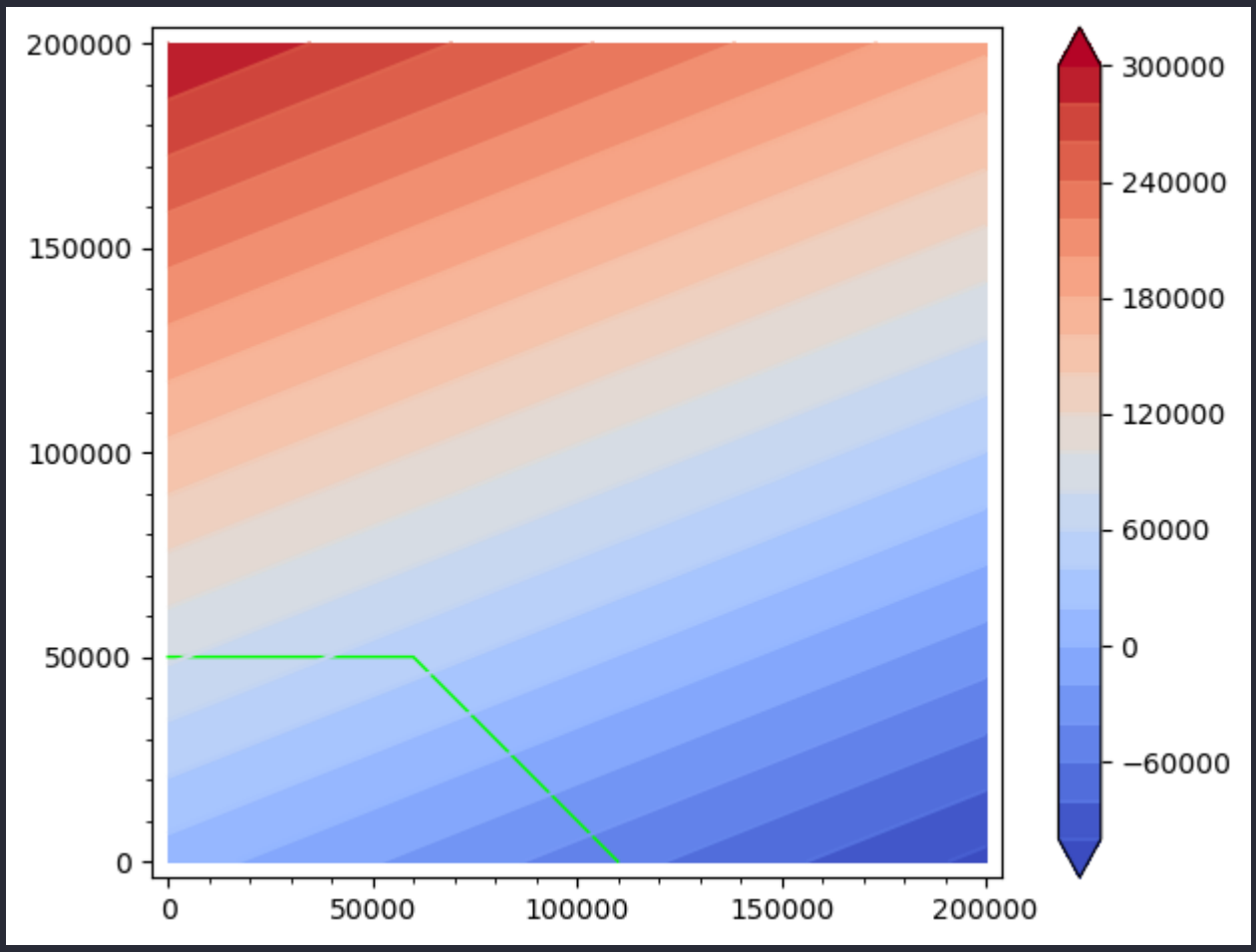

Figure 1: Contour plot of the profit function with the linear constraint function in green.

```python
# Show that the pig is gaining weight at 5 lbs / day at t = 0:
w(t) = 800 / (1 + 3 * (math.e)^(-t/30))
w_prime(t) = w.derivative(t)
w_pprime(t) = w_prime.derivative(t)
w_ppprime(t) = w_pprime.derivative(t)

print(f"At t=0, the pig is gaining {w_prime(0)} lbs per day")

# Calculate as t gets large
# t_0 = 3 # bad guess; close to crit point
t_0 = 30
guesses=[t_0]

for i in range(10):
    t_0 = t_0 - w_pprime(t_0) / w_ppprime(t_0)
    guesses.append(t_0)

root = round(guesses[-1], 2)
print(f'There is a root at {root}. It corresponds to maximum growth rate of {round(w_prime(root),3)}.')


# Returns:
# At t=0, the pig is gaining 5.0 lbs per day
# There is a root at 32.96. It corresponds to maximum growth rate of 6.667.
```

Listing 2: Exercise 2 (a) Code

```
# new pig problem
t = var('t')  # time in days

# Parameters
w0 = 200   # init weight of pig
rw = 5     # growth rate of pig (lb/day)
m0 = 0.65  # init market price ($/pound)
rm = -0.01 # market price rate of change ($/lbs/day)
d  = 0.45  # daily cost to keep the pig

# Functions
w(t) = 800 / (1 + 3 *(math.e)^(-t/30))   # weight of pig
m(t) = m0 + rm*t   # market price
r(t) = w(t)*m(t)   # revenue
c(t) = d*t         # cost
p(t) = r(t) - c(t) # profit

# Solve the Model
pprime    = p.derivative(t)        # derivative of profit
ppprime = pprime.derivative(t)

t_0 = 2
guesses=[t_0]
for i in range(10):
    t_0 = t_0 - pprime(t_0) / ppprime(t_0)
    guesses.append(t_0)

root = guesses[-1]
print(guesses)
print(f"There is a root at {root}. It corresponds to a profit of
↪  ${p(root)}.")
show(plot(pprime(t), 0, 20))
show(plot(p(t),0,20))

# Returns:
# There is a root at 12.33. It corresponds to a profit of $135.42.
```

Listing 3: Exercise 2 (b) Code

```
# Variables
x = var('x') # sub price
y = var('y') # ad price

# Parameters
subs_0 = 80_000  # current subscribers (subscriber / week)
ads_0  = 350     # current ads (pages / week)
d_p_ad  = 250    # dollars per advertisement
d_p_pap = 1.50   # dollars per paper (per subscriber)

slope_price_subs = -50000 # -5000 subs per $0.10
slope_price_ads = -0.5    # -50 pages per $100
slope_ads_subs = 20       # 20 subs per page

# Equations
A(y) = ads_0 + slope_price_ads * (y - d_p_ad)
S(x, y) = subs_0 + slope_price_subs * (x - d_p_pap) + slope_ads_subs * (A(y) - ads_0)
z(x, y) = (x * S(x, y)) + (y * A(y))

#first derivatives
F(x,y) = z.derivative(x)
G(x,y) = z.derivative(y)

print(f"{F=}")
print(f"{G=}")

#Second derivatives
F_x(x,y) = F.derivative(x)
F_y(x,y) = F.derivative(y)
G_x(x,y) = G.derivative(x)
G_y(x,y) = G.derivative(y)

print(f"{F_x=}")
print(f"{F_y=}")
print(f"{G_x=}")
print(f"{G_y=}")
```

Listing 4: Exercise 3 (b) Code

```python
OF(x,y) = [F(x,y), G(x,y)]


def df(v):
    (x,y) = v
    return Matrix([grad(j)(x,y) for j in OF])

X = vector([1.5, 350])

guesses=[X]
for i in range(5):
    (x,y) = X
    X = X - df(X).inverse() * OF(x,y)
    guesses.append(X)

print(guesses)
print(f'There is a solution at approximately {guesses[-1]}')
```

Listing 5: Exercise 3 (c)